Rishi Chilveri

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EDUCATION

Bharati Vidyapeeth (Deemed to Be University) **Navi Mumbai, Maharashtra**
Bachelor of Engineering in Computer Science — CGPA: 9.64 *2022 – 2026*

WORK EXPERIENCE

Machine Learning Intern – Model Development & Web Integration
Guardian AI *June 2026 – August 2026*

- Built a multi-label brand-safety classifier for 6 toxicity categories on 159K+ samples using TF-IDF and Logistic Regression, with precision-recall based threshold tuning.
- Developed an XGBoost CTR and conversion predictor integrated with FastAPI to forecast ad performance and flag underperforming creatives.
- Integrated Gemini, Replicate, and SERP-based market intelligence into an automated ad-generation pipeline with Streamlit and SQLite-based campaign tracking.

PROJECTS

Self-Healing RAG - LangGraph, ChromaDB, Groq, BM25, Cross-Encoder, Streamlit (2026)

* Engineered a self-healing Retrieval-Augmented Generation pipeline using LangGraph, deploying a critic agent that validates answer groundedness and triggers automatic query reformulation, achieving a 0% hallucination rate on a self-built 30-question evaluation set.
* Implemented hybrid retrieval combining BM25 keyword search and vector similarity via Reciprocal Rank Fusion, followed by cross-encoder reranking, improving answer accuracy from 92% (baseline RAG) to 96%.
* Built an automated evaluation harness benchmarking retrieval accuracy, hallucination rate, and latency across two RAG architectures, and deployed the system with multi-model fallback for uninterrupted inference under per-model rate limits.

Pixel Studio - React, FastAPI, ComfyUI, Stable Diffusion (2026)

* Engineered a full-stack text-to-image generator using Stable Diffusion, streaming results to a React frontend via FastAPI WebSocket.
* Achieved average generation time of 10–12s on consumer hardware with real-time progress updates to the UI.
* Applied attention slicing to optimize model execution on low-VRAM GPUs, reducing OOM crashes by 30%.

AI Marketing Agency - FastAPI, Streamlit, XGBoost, scikit-learn, Gemini API, Replicate API (2026)

* Developed an automated campaign platform that transforms a product idea into market insights, ad creatives, safety checks, performance predictions, and social-media publishing.
* Designed asynchronous FastAPI workflows with background job tracking and Streamlit UI for campaign creation, asset generation, and real-time status monitoring.
* Implemented a feedback loop using SQLite campaign outcomes to track generated ads and support future model improvement based on real campaign performance.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript
Frontend: React.js, HTML, CSS
Backend & Databases: FastAPI, Flask, Django, PostgreSQL, MongoDB, MySQL
ML & AI: PyTorch, TensorFlow, CNN, ANN, Stable Diffusion, InsightFace, FAISS, MediaPipe, Transfer Learning, Computer Vision
Tools & DevOps: Git, Docker, WebSocket, ComfyUI

ACHIEVEMENTS & CERTIFICATIONS

- **Convolutional Neural Networks** — DeepLearning.AI, authorized by Andrew Ng via Coursera (Jul 2025). Verify
- **Mastering Generative AI and ChatGPT** — 5-week course by GeeksforGeeks. Verify
- Passed the **Python 3.4.3 Test** conducted by IIT Bombay's Spoken Tutorial Project with a score of 70%, earning 4 credits. Verify